

Here are short questions and answers based on the list of numerical methods you provided, categorized by topic.

Root-Finding Methods

1. Iteration Method

- **Question:** What is the fundamental idea behind the Iteration Method for solving equations?
- **Answer:** The equation $f(x) = 0$ is rearranged into the form $x = g(x)$. An initial guess x_0 is then used to generate a sequence $x_1 = g(x_0)$, $x_2 = g(x_1)$, and so on, until the solution converges.

2. Bisection Method

- **Question:** What is the primary advantage of the Bisection Method?
- **Answer:** Its primary advantage is that it is guaranteed to converge to a root, provided the function is continuous on the initial interval $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs.

3. False Position Method (Regula Falsi)

- **Question:** How does the False Position Method improve upon the Bisection Method?
- **Answer:** Instead of using the midpoint, it uses the x-intercept of the secant line connecting the endpoints of the interval, which often leads to faster convergence.

10. Newton–Raphson Method

- **Question:** What is a significant drawback of the Newton-Raphson method?
- **Answer:** It requires the derivative of the function and can fail to converge if the initial guess is too far from the true root or if the derivative is zero at any point in the iteration.

Linear System Solving

4. Gauss Elimination Method

- **Question:** What is the end goal of the forward elimination phase in Gauss Elimination?
- **Answer:** To transform the system of linear equations into an upper triangular form.

5. Gauss Jordan Elimination

- **Question:** How does the Gauss-Jordan method differ from standard Gauss Elimination?
- **Answer:** It continues the process beyond an upper triangular matrix to achieve a reduced row echelon form (RREF), where the solution is directly readable without back-substitution.

9. Cramer's Rule

- **Question:** What is the major practical limitation of Cramer's Rule?
 - **Answer:** It is computationally very inefficient for large systems of equations, as it requires calculating the determinant of $n+1$ matrices for an $n \times n$ system.
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Regression & Curve Fitting

6. Least Squares Regression

- **Question:** What is the objective of the Least Squares Regression method?
- **Answer:** To find the line (or curve) that minimizes the sum of the squares of the vertical distances (residuals) between the observed data points and the predicted values from the model.

7. Logistic Regression

- **Question:** What type of problems is Logistic Regression primarily used for?
- **Answer:** It is used for classification problems, especially binary classification, to predict the probability that an observation belongs to one of two categories.

20. Polynomial Regression

- **Question:** When would you use Polynomial Regression over Simple Linear Regression?
 - **Answer:** When the relationship between the independent and dependent variables is clearly non-linear and can be better captured by a curve (polynomial) rather than a straight line.
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Optimization

8. Stochastic Gradient Descent

- **Question:** How does Stochastic Gradient Descent (SGD) differ from standard Gradient Descent?
 - **Answer:** Instead of using the entire dataset to compute the gradient for each update (which is slow), SGD uses a single, randomly selected data point or a small mini-batch, making it much faster for large-scale problems.
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Interpolation & Approximation

11. Interpolation — Diagonal & Horizontal Differences

- **Question:** What do the "Diagonal Differences" in interpolation typically refer to?
- **Answer:** They refer to the construction of a difference table, where diagonal entries are used in Newton's Forward or Backward Interpolation formulae.

12. Lagrange Interpolation Method

- **Question:** What is a key advantage of the Lagrange Interpolation method?
 - **Answer:** It can be applied even when the data points are not equally spaced.
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Numerical Integration

13. Trapezoidal Rule

- **Question:** What geometric shape is used to approximate the area under the curve in the Trapezoidal Rule?
- **Answer:** A series of trapezoids.

14. Simpson's 1/3 Rule

- **Question:** What is the fundamental requirement for applying Simpson's 1/3 Rule?
- **Answer:** The number of subintervals must be even, and it approximates the function over every two subintervals with a parabola.

15. Simpson's 3/8 Rule

- **Question:** What curve does Simpson's 3/8 Rule use to approximate the integral over three subintervals?
 - **Answer:** A cubic polynomial (a third-degree polynomial).
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Differential Equation Solvers

16. Euler's Method

- **Question:** What is the main limitation of Euler's Method for solving ordinary differential equations (ODEs)?
- **Answer:** It is a first-order method, meaning it can have significant truncation error unless the step size is very small, making it less accurate than higher-order methods.

17. Runge-Kutta Method

- **Question:** Why is the 4th-order Runge-Kutta (RK4) method so popular?
- **Answer:** It offers a good balance between computational effort and accuracy, providing fourth-order accuracy without requiring the calculation of higher-order derivatives.

18. Milne's Method

- **Question:** What is a defining characteristic of Milne's Method?
- **Answer:** It is a multi-step predictor-corrector method, meaning it uses information from several previous points to predict and then correct a new value.

19. Picard's Method

- **Question:** Picard's method is an iterative technique for solving what?
- **Answer:** It is used to solve initial value problems for ordinary differential equations by successively approximating the solution through integration.